

Position Terrain / Topological Alpha Workstation: A Gauge-Audited Algorithm and Software Architecture

Executive summary

A viable synthesis of the ideas developed across the prior work is possible, but the product should be designed around a stricter proposition than “topology finds alpha.”

The strongest form is a **Position-Conditioned Topological Sheaf-Fourier Terrain system**, abbreviated here as **PCT-SFT**:

$(\text{market history, position, strategy, horizon}) \longrightarrow [\text{conditional outcome terrain, historical analogs, uncertainty, ev}]$

Its defining feature is that an arbitrary options position is treated as a **time-dependent query against a typed market-state geometry**, rather than as a static payoff diagram or an indicator-triggered trade.

For a position P , strategy π , and horizon h , the central object is

$$\widehat{\mathcal{L}}(Y_{t,h}(P, \pi) \mid Z_t(P, \pi)),$$

where $Y_{t,h}$ is a distribution of future position outcomes—PnL, maximum adverse/favorable excursion, drawdown, barrier touches, Greek migration, costs, and other declared quantities—and Z_t is a joint encoding of the current position, slower market condition, faster market activity, topological/geometric state, strategy state, and explicitly typed time coordinates.

The mathematical stack has a sound research basis. Graph signal processing treats Laplacian eigenvectors as graph-adapted spectral bases; Singer and Wu extend the idea to vector-valued data through connection-Laplacian geometry and vector diffusion; Hansen and Ghrist lift Laplacian methods to cellular sheaves; and a very recent August 2026 preprint develops a unified **Sheaf Fourier Transform**, filtering, and sampling framework for heterogeneous network signals. ¹

Persistent homology is useful as a complementary descriptor rather than a substitute for statistics. Perea and Harer establish a theory connecting sliding-window embeddings and persistent topology of time series, while Gakhar and Perea show that quasiperiodic signals with rationally independent frequencies can generate sliding-window clouds dense in higher-dimensional tori. These results justify using TDA to characterize state trajectories, but they also illustrate why visually rich topology can be induced by the representation itself. ²

That epistemic distinction is already central to the uploaded commercialization blueprint: the product should distinguish the geometry inserted by an analyst from exact consequences of that representation,

numerical certificates, descriptive observations, validated findings, and speculative interpretations. [filecite turn0file1](#) The same principle should govern the market product.

The uploaded neuroevolution/options analysis adds the essential financial correction: unlike a videogame environment where the controller can experience many repeatable worlds, historical markets provide a limited, nonstationary realized history. Consequently, increasing the number of strategy or model variants can increase selection bias faster than it increases knowledge. [filecite turn0file2](#) The original Mario architecture is nevertheless a useful software metaphor: convert the environment into a structured numerical state, let a model react to that state rather than memorize a fixed script, and test generalization on environments not used to select the model. [filecite turn0file0](#)

The market workstation should therefore be constructed as a **baseline-first scientific instrument**. Connection/sheaf spectra, TDA, neural models, and mixture-of-experts routing are candidate feature layers; none receives production status until it improves a frozen held-out task over simpler alternatives. The operational visualization and specialist-routing concepts in the uploaded GT-MoE deck are useful as interface and systems-design inspiration, but their geometric claims should remain hypotheses until separately validated. [filecite turn0file3](#)

The recommended algorithm is a hybrid rather than a monolithic network:

Point-in-time event normalization

- typed multiscale state
- causal graph + connection
- connection/sheaf spectral descriptors
- +persistent-topology descriptors
- +position transport + position encoding
- +strategy automaton + microstructure encoding
- analog terrain
- distributional model heads
- evidence/V&V ledger.

The most important architectural separation is between **three different kinds of transport**:

1. **Position transport** maps an economically comparable option position from one historical option-chain fiber to another.
2. **Connection transport** maps local coordinate frames between market-state vertices.
3. **Strategy evolution** moves an execution policy through its deterministic state machine.

They must not be conflated.

The most important mathematical safeguard is a **gauge-fair comparator**. On a chronological tree, orthogonal transports can be gauge-unwrapped; therefore a connection basis must be compared not only with a raw DCT but with a DCT or scalar graph Fourier basis applied after the same frame alignment. Synchronization theory identifies cycle holonomy—not arbitrary coordinate rotation along a tree—as the gauge-invariant obstruction to global alignment. [3](#)

The most important statistical safeguard is that the generalization unit must be a **world, regime block, instrument-period, or independent deployment period**, not an individual bar. The previously discussed Gate E1 simulation is consequently best retained as an oracle implementation certificate. It does not yet validate market compression because the synthetic signals were constructed from the same low connection-Laplacian modes used to reconstruct them.

The resulting V&V ladder should be:

E1 representation → *E2 holonomy* → *E3 identifiability* → *E4 predictive utility* → *E5 live robustness*

with an operator-sanity E0 prerequisite. Only E4 can begin to support a narrowly defined claim of predictive or trading utility. Bailey and López de Prado's Deflated Sharpe Ratio and Bailey et al.'s Probability of Backtest Overfitting provide useful controls once the workstation begins searching strategy/model families rather than merely characterizing representations. 4

Finally, the local LFM layer should function as a **semantic compiler and analyst interface, never as numerical authority**. Current official LiquidAI model repositories include LFM2.5 GGUF variants intended for local/on-device deployment; for example, the 1.2B Instruct repository lists a Q4_K_M file of roughly 731 MB, while an 8B-A1B variant lists roughly 5.16 GB at Q4_K_M. Exact model suitability and licensing should nonetheless be treated as unresolved project assumptions until reviewed, and every reproducible experiment should pin the exact repository revision, tokenizer, quantization, and checksum rather than referring to "latest." 5

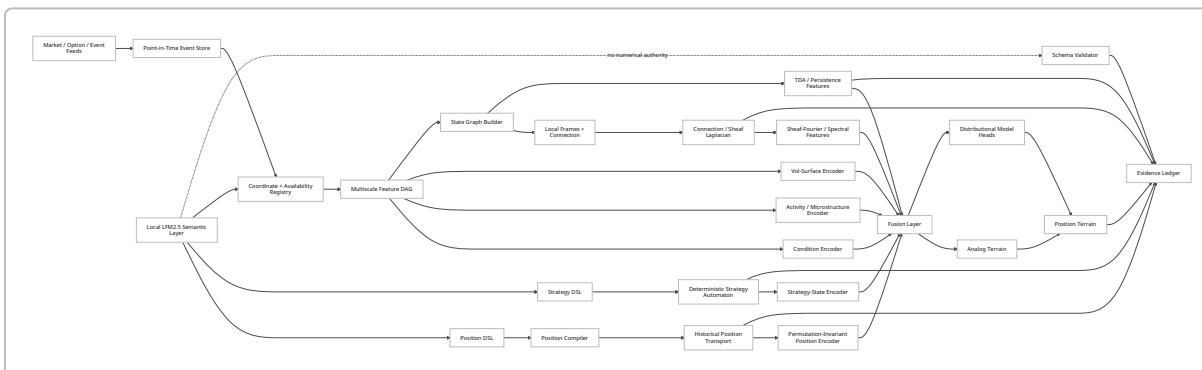
The overall research judgment is therefore:

Proceed, but define the product as a gauge-audited, position-conditioned market-state laboratory whose advanced representations must earn their complexity through frozen validation.

That is both mathematically defensible and commercially differentiated.

Algorithmic architecture and data model

The proposed PCT-SFT architecture has four conceptual planes: the **event plane**, **mathematical representation plane**, **position/policy plane**, and **evidence plane**.



Canonical event model

Live mode and historical playback should consume the same event type:

$$e = (t_{\text{event}}, t_{\text{ingest}}, t_{\text{available}}, \text{source}, \text{instrument}, \text{contract}, \text{event type}, \text{revision}, \text{payload}, \text{quality}).$$

The critical field for research is $t_{\text{available}}$: the earliest instant at which the information could legally enter the simulated decision process. A backtest should use that field rather than retrospectively cleaned timestamps.

This event-sourced design makes live and replay operation differ only in their clock:

```

historical event log + virtual clock
      or
live feed + wall/event clock
      ↓
identical state pipeline

```

The prior GT-MoE concept's synchronized streaming, geometry, topology, model activation, and visualization structure translates well into this architecture, provided the mathematical claims remain separately audited. [filecite:turn0file3](#)

Coordinate registry

Time and position coordinates need explicit semantics. The scientific-studio blueprint correctly identifies coordinate confusion as an important source of false interpretation. [filecite:turn0file1](#)

Coordinate	Type	Example	Must not be confused with
Event time	Physical/exchange time	09:31:02.500 ET	Ingestion time
Ingestion time	System time	Feed arrival	Market event time
Availability time	Causal information time	After revision/ processing	Original publication timestamp
Replay clock	Simulation coordinate	10× historical speed	Physical market time
Forecast horizon h	Outcome coordinate	30 min, 5 sessions	Option tenor
Option tenor τ	Contract coordinate	21 DTE	Forecast horizon
Log-moneyness	Position coordinate	$\log(K/F_t)$	Raw strike
Delta	Derived contract coordinate	0.25 delta	Contract identity
Filtration ϵ	TDA scale	Rips threshold	Time

Coordinate	Type	Example	Must not be confused with
Diffusion parameter	Operator scale	e^{-tL}	Exchange time
Strategy age	Policy-state coordinate	bars since trigger	Contract age

Every generated artifact should preserve this registry.

Market state decomposition

At each market time t ,

$$M_t = (C_t, A_t, \Sigma_t, E_t),$$

where:

- C_t is the slower **condition state**;
- A_t is the faster **activity/microstructure state**;
- Σ_t is the option-volatility surface state;
- E_t contains timestamped semantic/event context.

A canonical state vector can be written

$$x_t = x_t^{\text{price}} \oplus x_t^{\text{technical}} \oplus x_t^{\text{micro}} \oplus x_t^{\text{surface}} \oplus x_t^{\text{calendar}} \oplus x_t^{\text{semantic}}.$$

Technical indicators such as RSI, MACD, moving-average distances, ATR-like measures, or volume oscillators should be treated as a large **coordinate dictionary**, not as independent sources of truth. Redundant indicators will often encode overlapping transformations of the same price/volume history. Their incremental information must therefore be tested by family-level ablation rather than inferred from the number of indicator names.

For the fast activity tower, established microstructure work provides sensible primitives. Cont, Kukanov, and Stoikov found short-horizon price changes strongly related to order-flow imbalance relative to available depth, while Cont, Stoikov, and Talreja model the probabilities of quote movements and executions conditional on limit-order-book state. ⁶ DeepLOB supplies an example of using separate spatial and temporal neural mechanisms for limit-order-book data, but it should be a learned comparator rather than the first baseline. ⁷

A simple order-flow imbalance primitive is

$$\text{OFI}_t = \Delta V_t^b - \Delta V_t^a,$$

augmented by spread, depth, queue depletion, trade arrival intensity, cancel intensity, microprice, and quote age. The exact definitions should be versioned because seemingly small feed conventions can materially change the feature.

Arbitrary position representation

Represent a position as an unordered set of legs

$$P_t = \{\ell_{j,t}\}_{j=1}^L,$$

with

$$\ell_{j,t} = (u_j, K_j, T_j, c_j, q_j, m_j, \text{market quote, derived Greeks, liquidity}).$$

A normalized leg coordinate can be

$$\xi_{j,t} = \left[\log \frac{K_j}{F_t}, \tau_{j,t}, \mathbf{1}_{\text{call}}, q_j m_j, \sigma_{j,t}, \Delta_{j,t}, \Gamma_{j,t}, \nu_{j,t}, \Theta_{j,t}, \text{spread}_{j,t} \right].$$

Leg order should not matter. A Deep-Sets-style encoder therefore fits the mathematical symmetry:

$$E_P(P_t) = \rho \left(\sum_{j=1}^L \phi(\xi_{j,t}) \right).$$

Deep Sets formalizes neural architectures for permutation-invariant functions over sets; a later attention-based implementation could substitute a Set Transformer if interactions among legs justify the added complexity. ⁸

The learned encoder should always be accompanied by deterministic position summaries:

$$g_t(P) = [\Delta_P, \Gamma_P, \nu_P, \Theta_P, \rho_P, \text{debit/credit, scenario PnL, liquidity burden}].$$

Those deterministic quantities remain visible even if the learned encoder is disabled.

Position transport

A current option contract usually has no literal historical counterpart. Let

$$\mathcal{F}_t$$

denote the fiber of available contracts at time t . The workstation therefore needs an explicit economic transport

$$T_{t \rightarrow u}^P : \mathcal{F}_t \rightarrow \mathcal{F}_u.$$

This is **not** the same object as the connection transport used later to align market-state frames.

Three transport modes should be implemented.

Contract replay follows actual contract identifiers over their real lifetime and uses no synthetic cross-era mapping.

Template transport replaces a leg with the closest historical contract under a declared cost such as

$$d_{\Theta}^2(j, c) = w_m \left(\log \frac{K_j}{F_t} - \log \frac{K_c}{F_u} \right)^2 + w_{\tau} (\tau_j - \tau_c)^2 + w_{\Delta} (\Delta_j - \Delta_c)^2 + w_L L_c.$$

Exposure transport is more powerful. Let A_u map historical contract quantities to an exposure vector containing selected Greeks and scenario payoffs. Solve

$$q_u^* = \arg \min_q \|A_u q - \tilde{e}_{t \rightarrow u}(P)\|_W^2 + \lambda \|q\|_1 + \eta C_u(q),$$

subject to declared contract-type, expiration, liquidity, and position-size constraints.

Here $\tilde{e}_{t \rightarrow u}(P)$ is the desired transported economic exposure, not a historical answer selected after observing future PnL.

Each transport must persist its residual:

$$r_{\text{transport}} = \|A_u q_u^* - \tilde{e}\|_W,$$

so poor historical analogs can be rejected rather than disguised.

Strategy grammar

A strategy is compiled into a deterministic automaton:

UNSEEN $\rightarrow$ SCANNED $\rightarrow$ QUALIFIED $\rightarrow$ FORMING $\rightarrow$ ELIGIBLE $\rightarrow$ TRIGGERED $\rightarrow$ MANAGED $\rightarrow$ {EXITED, INVALIDATED,

This turns human language such as “take the first pullback above VWAP when relative volume is elevated” into individually inspectable predicates.

The local language model may generate the typed DSL, but the deterministic replay engine decides whether each transition occurred.

Fusion and model heads

The joint position-state representation is

$$Z_t(P, \pi) = E_C(C_t) \oplus E_A(A_t) \oplus E_{\Sigma}(\Sigma_t) \oplus E_G(G_t) \oplus E_{\text{TDA}}(t) \oplus E_P(P_t) \oplus E_{\pi}(S_t^{\pi}).$$

The first implementation should concatenate these blocks and use family-level masks so every tower can be removed independently.

A later gated model may use

$$h_t = \sum_{g=1}^G \alpha_g(P_t, C_t) h_t^{(g)}, \quad \sum_g \alpha_g = 1,$$

with α_g learned only after the simpler concatenated model passes validation.

This is preferable to assigning literal semantic meanings to individual hidden neurons. An indicator can have a dedicated input coordinate or tower, but a hidden neuron should not be labeled “the MACD neuron” unless controlled interventions demonstrate that relationship.

Distributional heads should predict quantities such as

$$\hat{Q}_{0.05}, \hat{Q}_{0.25}, \hat{Q}_{0.50}, \hat{Q}_{0.75}, \hat{Q}_{0.95}$$

for position PnL, along with separate event probabilities for stop touch, target touch, large drawdown, or liquidity impairment. Conformalized quantile regression is one possible calibration layer, but its classical finite-sample guarantee relies on exchangeability; a market implementation would therefore need time-aware calibration rather than assuming ordinary iid coverage. ⁹

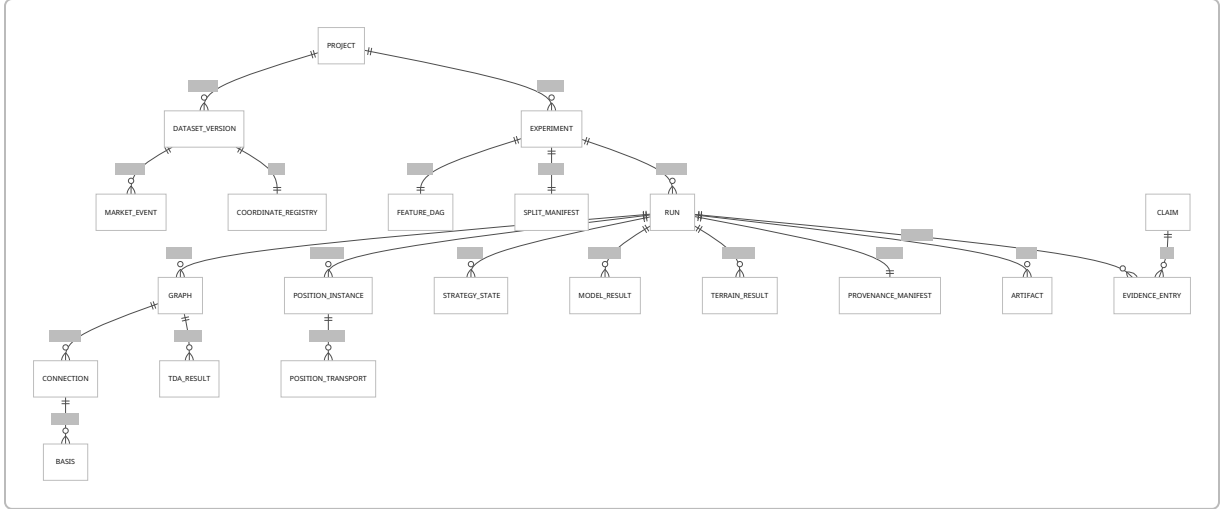
Evidence ledger

Every result should link to an evidence state such as:

Status	Meaning
EXACT	Mathematical consequence of declared definitions
CERTIFIED	Numerical invariant or implementation identity passed
DESCRIPTIVE	Observed in current data
VALIDATED	Survived frozen holdout/null design
UNSUPPORTED	Declared positive claim failed its test
INCONCLUSIVE	Insufficient discrimination or power
FAILED_CHECK	Numerical/software integrity gate failed
SPECULATIVE	Interpretation not yet tested

That evidence architecture follows the uploaded scientific-studio blueprint and should be one of the core commercial differentiators. fileciteturn0file1

A minimal entity graph is:



Mathematical core

Connection construction

Let

$$G = (V, E, w)$$

be a graph of market states or eras. For the MVP, attach a common r -dimensional stalk

$$\mathcal{F}_i = \mathbb{R}^r$$

to every vertex.

For oriented edge $i \rightarrow j$, let

$$U_{ij} \in O(r)$$

transport local coordinates from the frame at i to the frame at j , with

$$U_{ji} = U_{ij}^\top.$$

A section is

$$s = (s_1, \dots, s_n), \quad s_i \in \mathbb{R}^r.$$

Its connection smoothness energy is

$$\mathcal{E}_U(s) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} \|s_j - U_{ij}s_i\|_2^2.$$

The corresponding connection Laplacian satisfies

$$\mathcal{E}_U(s) = \frac{1}{2} s^\top L_U s.$$

With the stated orientation convention,

$$(L_U)_{ii} = d_i I_r, \quad d_i = \sum_j w_{ij},$$

and

$$(L_U)_{ij} = -w_{ij} U_{ij}^\top, \quad (L_U)_{ji} = -w_{ij} U_{ij}.$$

This is the discrete vector-valued geometry underlying connection-Laplacian methods and $O(d)$ synchronization. ¹⁰

A normalized form is

$$\mathcal{L}_U = D^{-1/2} L_U D^{-1/2}, \quad D = \text{diag}(d_i I_r).$$

The normalized and unnormalized variants answer slightly different questions; the run manifest must state which is used.

General cellular-sheaf form

For a genuine cellular sheaf, vertex and edge stalks may be heterogeneous and restrictions need not be orthogonal.

For an edge $e = (i, j)$, define

$$\rho_{i \rightarrow e} : \mathcal{F}_i \rightarrow \mathcal{F}_e, \quad \rho_{j \rightarrow e} : \mathcal{F}_j \rightarrow \mathcal{F}_e.$$

Then the coboundary is

$$(\delta s)_e = \rho_{j \rightarrow e} s_j - \rho_{i \rightarrow e} s_i,$$

and

$$L_{\mathcal{F}} = \delta^\top \delta.$$

Hansen and Ghrist develop the relationship between these sheaf Laplacians, cellular structure, and sheaf cohomology. ¹¹

The workstation must maintain a hard boundary between:

graph connection / validated higher cellular sheaf.

Pairwise transports are not, by themselves, proof that compatible higher-cell restriction maps exist; the uploaded blueprint explicitly makes that compatibility gate part of the scientific design.

filecite turn0file1

Sheaf Fourier transform

Diagonalize the symmetric connection or sheaf Laplacian:

$$L_{\mathcal{F}} = \Psi \Lambda \Psi^{\top},$$

where

$$0 \leq \lambda_1 \leq \lambda_2 \leq \dots.$$

For a section s ,

$$\hat{s} = \Psi^{\top} s,$$

and reconstruction using the first k modes is

$$\tilde{s}_k = \Psi_k \Psi_k^{\top} s.$$

Low eigenvalues correspond to sections with low sheaf inconsistency energy. The 2026 D'Acunto–Di Nino–Di Lorenzo–Barbarossa preprint explicitly develops this into a Sheaf Fourier Transform and extends filtering and sampling to heterogeneous network signals. It is highly relevant to this workstation, but it remains a recent preprint rather than established market validation. ¹²

Individual eigenvectors cease to be canonical when eigenvalues are repeated. For learned features, the workstation should therefore support eigenspace/projector statistics such as

$$P_B = \Psi_B \Psi_B^{\top}$$

for an eigenvalue band B , and spectral energy

$$E_B(s) = \|P_B s\|_2^2.$$

These remain stable under arbitrary basis rotations within a degenerate eigenspace.

Gauge transformations

Let each vertex choose a new local orthogonal basis

$$G_i \in O(r),$$

and let

$$G = \text{diag}(G_1, \dots, G_n).$$

Then

$$s'_i = G_i s_i,$$

and

$$U'_{ij} = G_j U_{ij} G_i^\top.$$

The Laplacian transforms by orthogonal conjugation:

$$L'_U = G L_U G^\top.$$

Therefore its eigenvalues and other conjugacy-invariant spectral quantities are unchanged. This is why random-gauge tests should be automatic numerical certificates. Connection/synchronization theory makes this gauge structure explicit. ³

Gauge-fair DCT and graph-Fourier comparators

On a spanning tree, choose a root r with

$$G_r = I.$$

For each tree edge $i \rightarrow j$, recursively set

$$G_j = G_i U_{ij}^\top.$$

Then

$$G_j U_{ij} G_i^\top = I$$

for every tree edge.

Consequently, tree-edge connection smoothness becomes ordinary scalar/vector smoothness in the unwrapped coordinates.

This means the meaningful E1 comparator is not merely:

connection SFT versus raw DCT.

It must include:

connection SFT versus gauge-unwrapped DCT/GFT.

Only non-tree edges can retain irreducible cycle inconsistency. Synchronization geometry identifies cycle holonomy as the obstruction to global trivialization. ¹³

Holonomy

For a cycle

$$c = (v_0, v_1, \dots, v_m = v_0),$$

define

$$H_c = U_{v_{m-1}v_m} \cdots U_{v_1v_2} U_{v_0v_1}.$$

Under a gauge change,

$$H'_c = G_{v_0} H_c G_{v_0}^\top,$$

so its conjugacy class is gauge invariant.

For $SO(2)$,

$$H_c = R(\phi_c)$$

and the holonomy angle can be recovered as

$$\phi_c = \text{atan2}((H_c)_{21}, (H_c)_{11}).$$

Equivalent inconsistency statistics include

$$2 - \text{tr}(H_c) = 2(1 - \cos \phi_c)$$

and

$$\|I - H_c\|_F^2 = 4(1 - \cos \phi_c).$$

For higher $O(r)$, use eigenangles, singular/conjugacy invariants, trace-based distances, and determinant as an orientation indicator. Note that

$$\det H_c = 1$$

for $SO(r)$, so determinant alone cannot detect rotational holonomy in that group.

Recent work on twisted Laplacians provides additional mathematical evidence that cycle holonomy controls spectral frustration and the existence or absence of zero modes in such systems. ¹⁴

This should still be described as **discrete connection holonomy**, not market curvature in a physical sense.

Rate-distortion accounting

Gate E1 should measure actual compression rather than merely number of retained coefficients.

For basis B and reconstruction rank k ,

$$D_B(k) = \frac{1}{N} \|x - \tilde{x}_{B,k}\|_2^2.$$

But matched k is not matched bitrate. Define

$$R_B = R_{\text{coeff}} + \frac{R_{\text{graph}} + R_{\text{transport}} + R_{\text{basis}} + R_{\text{metadata}}}{N_{\text{amortized signals}}}.$$

If the decoder deterministically regenerates Ψ from the graph and transports, the eigenbasis itself need not be stored, but graph structure, weights, transport parameters, solver/version information, and coefficient quantization still constitute side information.

Graph-based transform coding literature already evaluates Laplacian-derived transforms through rate-distortion rather than retained dimension alone, which is the correct framing for E1. 15

Gate E1 should therefore contain both:

$$D(k)$$

and

$$D(R)$$

curves.

Compression versus denoising

Suppose the synthetic generator is

$$x = s + \varepsilon.$$

Two different targets are required.

Observed-signal reconstruction:

$$D_{\text{obs}} = \frac{1}{N} \|x - \tilde{x}\|^2.$$

Latent denoising:

$$D_{\text{clean}} = \frac{1}{N} \|s - \tilde{x}\|^2.$$

The first asks how compactly the observed sample can be represented. The second asks whether the basis removes known injected noise.

Real market data do not reveal a clean latent s , so only the first is directly observable without an additional generative assumption.

Persistent topology

For each market time t , construct a sliding-window cloud

$$X_t = \{x_{t-m+1}, \dots, x_t\},$$

or the corresponding spectral state points.

A Vietoris–Rips complex at filtration scale ϵ contains a simplex whenever all pairwise distances between its vertices are at most ϵ . Persistent homology then produces interval or diagram representations for $H_0, H_1, \dots$

Perea and Harer provide the theoretical basis for using sliding-window persistence to characterize periodic structure in time series. ¹⁶ Gakhar and Perea show why quasiperiodic multi-frequency trajectories can have dense torus closures. ¹⁷

The workstation should therefore distinguish:

trajectory dimension, local intrinsic dimension, orbit-closure dimension, persistent topology.

They are not interchangeable.

TDA features should initially enter the model only as a candidate family:

$$E_{\text{TDA}} = [\text{Betti curves, persistence statistics, landscape/image features, topological change score}].$$

E4 must decide whether they add predictive value beyond raw and spectral baselines.

Position-conditioned geometry

The most novel market-specific extension is to make state similarity depend partly on the position being evaluated.

Let h_t be a canonical market-state embedding, and let

$$e(P_t) = [|\Delta|, |\Gamma|, |\nu|, |\Theta|, \dots]$$

be normalized exposure weights.

Define a deterministic first-generation metric

$$d_P^2(t, u) = (h_t - h_u)^\top W_P (h_t - h_u) + \lambda_T d_{\text{transport}}^2(P_t, T_{t \rightarrow u}^P(P_t)),$$

with

$$W_P = W_0 + \sum_g \alpha_g(P) W_g.$$

A strongly vega-sensitive position can therefore weight surface-state differences more heavily than an almost pure-delta position, while a gamma-heavy short-dated position can give greater weight to fast activity and realized-volatility coordinates.

The weights should begin as transparent deterministic functions of exposures. Learned metric conditioning is a later E4 experiment.

This gives the terrain its core interpretation:

Two historical moments may be close for one position and far apart for another.

Concrete algorithms and pseudocode

The following algorithms define a practical first implementation rather than merely a conceptual architecture.

Causal graph and connection-basis construction

The graph should contain chronological edges plus carefully controlled similarity edges. Similarity edges may use only information available at the vertex's point-in-time cutoff.

```
ALGORITHM BuildConnectionBasis(states, rank  $r$ , neighbors  $k$ , modes  $m$ )
```

```
INPUT:
```

```
states[i]           point-in-time feature windows
available_at[i]      causal availability cutoff
construct_features   feature families allowed for graph construction
 $r$                  local frame rank
 $k$                  similarity-neighbor count
 $m$                  requested low modes
```

1. For every state i :
 $x_i = \text{causal_features}(\text{states}[i], \text{construct_features})$
2. Fit normalization ONLY on development data.
 Transform all admissible development states with frozen scaler.
3. Build graph:
 add chronological edges ($i, i+1$)
 add mutual-kNN similarity edges using only allowed comparisons
 record edge reason and distance
4. For every vertex i :
 gather local neighborhood N_i
 center local observations
 $Q_i = \text{top-}r \text{ left/right singular vectors from local PCA/SVD}$
 record singular values and local-rank diagnostics
5. For every edge (i, j):
 $M_{ij} = Q_j^T Q_i$
 $[A, \Sigma, B^T] = \text{SVD}(M_{ij})$
 $U_{ij} = A B^T$
 $\text{confidence}_{ij} = \text{function}(\Sigma, \text{neighborhood overlap, sample count})$

 if $\text{group} == SO(r)$ and $\det(U_{ij}) < 0$:

```

        apply declared orientation correction

        w_ij = similarity_weight(i,j) * confidence_ij

6. Assemble sparse block connection Laplacian L_U.

7. Run certificates:
    Hermitian/symmetric
    PSD within tolerance
    U_ji = U_ij^T
    random-gauge spectral invariance

8. Compute lowest m eigenpairs:
    L_U Psi = Psi Lambda

9. Persist:
    graph
    frames
    transports
    weights
    L_U
    eigenvalues/eigenspace projectors
    construction manifest

OUTPUT:
    ConnectionBasisBundle

```

Orthogonal Procrustes alignment and connection-Laplacian constructions follow the same broad alignment/spectral logic developed in vector diffusion maps and synchronization research. ¹⁰

A crucial implementation detail is that graph construction and transport estimation have their own **feature allowlist**. E3 can then test whether the resulting structure generalizes to a completely held-out feature family.

Gauge-unwrapping algorithm

```

ALGORITHM GaugeUnwrap(graph G, transports U)

1. Select a deterministic root r.
2. Compute deterministic spanning tree T of G.
3. Set gauge[r] = I.

4. Traverse T from root:
    for oriented tree edge parent i -> child j:
        gauge[j] = gauge[i] @ U[i,j].T

```

5. For every tree edge:


```

      U_prime[i,j] =
          gauge[j] @ U[i,j] @ gauge[i].T

      assert U_prime[i,j] approximately I
      
```
6. For every section s:


```

      y_i = gauge[i] @ s_i
      
```
7. Preserve all non-tree transformed edges:


```

      U_prime_non_tree[i,j]
      
```
8. Compute:


```

      raw-DCT reconstruction on original coordinates
      gauge-DCT reconstruction on y
      scalar-GFT reconstruction on y using same graph weights
      
```
9. Save tree, root, gauges, residuals and comparator results.

The non-tree transformed transports contain the cycle inconsistency that cannot be removed by tree alignment. This is the comparator required to prevent raw coordinate misalignment from masquerading as topological advantage. ¹³

Connection-transport estimation

For local PCA frames Q_i, Q_j ,

$$M_{ij} = Q_j^\top Q_i.$$

Take

$$M_{ij} = A \Sigma B^\top$$

and define

$$U_{ij} = AB^\top.$$

A useful edge confidence is

$$c_{ij} = \frac{1}{r} \text{tr}(\Sigma),$$

possibly multiplied by independent neighborhood-overlap and sampling-density factors.

Do not force a transport onto a low-confidence edge. Removing an edge is preferable to manufacturing an apparently precise orthogonal map.

A more robust production implementation should estimate transport uncertainty by bootstrap:

```

for b in bootstrap neighborhood resamples:
    estimate  $Q_i^{(b)}$ ,  $Q_j^{(b)}$ 
    compute  $U_{ij}^{(b)}$ 

summarize:
    mean alignment
    principal-angle spread
    holonomy uncertainty contribution

```

The bootstrap output should later enter Gate E2's null model.

Position-transport algorithm

ALGORITHM TransportPosition(template P at source t, target option chain C_u)

1. Convert source position to canonical exposure template:

- moneyiness coordinates
- tenor coordinates
- option type
- scenario payoff vector
- selected Greeks
- gross/net quantity constraints

2. Filter target chain C_u :

- information available at target timestamp only
- matching underlying
- allowed expirations
- quote quality
- liquidity thresholds

3. Build target exposure matrix A_u .

4. Solve:

$$q^* = \underset{q}{\operatorname{argmin}} \quad \begin{aligned} & ||A_u q - \text{target_exposure}||_W^2 \\ & + \lambda_{l1} ||q||_1 \\ & + \eta * \text{execution_cost}(q) \end{aligned}$$

subject to declared constraints.

5. Calculate:

- transport residual
- liquidity penalty
- achieved Greek mismatch
- scenario-payoff mismatch

6. If residual exceeds frozen threshold:
 mark analog INVALID_POSITION_TRANSPORT
 do not silently interpolate.
7. Persist:
 target contracts
 solver settings
 constraints
 objective components
 transport quality

The threshold belongs in the frozen experiment manifest rather than being tuned after inspecting the outcomes.

Spectral compression and denoising

```

ALGORITHM SpectralReconstruct(section x, basis Psi, rank k)

1. coefficients a = Psi[:, 1:k]^T x
2. optionally quantize a using declared codec
3. x_hat = Psi[:, 1:k] a
4. observed_MSE = mean((x - x_hat)^2)

5. if synthetic clean truth s is available:
    clean_MSE = mean((s - x_hat)^2)

6. compute coefficient storage bits
7. add amortized graph/transport/basis side information
8. return:
    reconstruction
    MSEs
    total rate
    spectral energy distribution

```

For market features, the same operation may be used as a denoising candidate, but it should not be described as noise removal until a validation experiment establishes that the discarded components are nuisance variation rather than predictive signal.

Terrain estimation

The first terrain estimator should be nonparametric and interpretable.

Let

$$z_t(P, \pi)$$

be the frozen joint representation.

For query z_* , find historical analogs $\mathcal{N}_k(z_*)$, excluding intervals that violate causal/purge rules.

Use kernel weights

$$w_i = \frac{K(d(z_*, z_i)/h)}{\sum_{j \in \mathcal{N}_k} K(d(z_*, z_j)/h)}.$$

Then the empirical outcome CDF is

$$\hat{F}_*(y) = \sum_{i \in \mathcal{N}_k} w_i \mathbf{1}(Y_i \leq y).$$

Weighted quantiles produce the terrain values.

ALGORITHM EstimateTerrain(query_state, position, strategy, horizon)

1. Compile position and verify current valuation.
2. Encode market state using frozen feature pipeline.
3. Encode position.
4. Encode strategy state.
5. Add validated geometry/TDA features only if their gate status allows them.
6. Find historical candidate states.
7. For each candidate u:
 - transport query position to chain at u
 - reject if transport residual too large
 - enforce purge/embargo and no-overlap rules
 - calculate joint position-conditioned distance
8. Retain nearest valid analogs.
9. Estimate weighted empirical distribution of:
 - net PnL
 - MFE
 - MAE
 - max drawdown
 - barrier events
 - Greek migration
 - costs
10. Calculate:
 - effective sample size
 - regime concentration
 - calendar concentration

transport residual distribution
 OOD distance

11. If support is insufficient:
 ABSTAIN / UNKNOWN

12. Return distribution + actual historical analog paths.

Effective weighted analog count can be reported as

$$N_{\text{eff}} = \frac{1}{\sum_i w_i^2}.$$

A low N_{eff} is not fixed by adding a neural model; it is evidence that the historical support is weak.

Distributional neural model

Only after the analog baseline is frozen should a neural residual or parallel head be evaluated.

For quantile set $\mathcal{Q}$,

$$\mathcal{L}_{\text{quantile}} = \sum_{\tau \in \mathcal{Q}} w_{\tau} \rho_{\tau}(Y - \hat{q}_{\tau}(Z)),$$

where

$$\rho_{\tau}(u) = u(\tau - \mathbf{1}_{u < 0}).$$

A mixture-of-experts variant is

$$p(Y|Z) = \sum_{m=1}^M \pi_m(Z) p_m(Y|Z),$$

with

$$\pi_m(Z) = \frac{e^{g_m(Z)}}{\sum_r e^{g_r(Z)}}.$$

The GT-MoE concept demonstrates the useful UI pattern of exposing gate weights and active experts rather than hiding routing internally. For markets, however, an MoE should be treated as a later comparator; a regime-specific expert that looks impressive in sample can simply add another layer of model selection.

Strategy-state replay

```
ALGORITHM ReplayStrategy(event_log, strategy_DSL, position_template)
```

```

state = UNSEEN
portfolio = EMPTY

for event e in chronological availability order:

    update canonical market snapshot
    update causal indicators
    update microstructure state
    update option chain
    update semantic-event state

    switch state:

        UNSEEN:
            if universe_filter(snapshot):
                state = SCANNED

        SCANNED:
            if qualification_rules(snapshot):
                state = QUALIFIED

        QUALIFIED:
            if setup_start(snapshot):
                state = FORMING

        FORMING:
            if setup_invalidated(snapshot):
                state = INVALIDATED
            elif trigger_eligibility(snapshot):
                state = ELIGIBLE

        ELIGIBLE:
            if trigger(snapshot):
                proposed_position =
                    instantiate(position_template, current_chain)

                fills =
                    execution_model(proposed_position, current_quotes)

                if fills valid:
                    portfolio = fills
                    state = TRIGGERED

        TRIGGERED:
            state = MANAGED

        MANAGED:
            mark position using declared accounting convention

```

```

        if hard_invalidation(snapshot):
            execute declared exit
            state = EXITED

        elif target_rule(snapshot):
            execute declared exit
            state = EXITED

        elif expiry_or_time_exit(snapshot):
            execute declared exit
            state = EXITED

    record every transition, rejected trigger,
    quote, fill assumption, mark and model output

```

The local LFM can turn language into the DSL, but the replay loop must remain ordinary deterministic software.

Local LFM semantic sidecar

The LFM should operate through a constrained contract:

```

User language
  ↓
Local LFM
  ↓
Typed JSON proposal
  ↓
JSON Schema / Pydantic validator
  ↓
Deterministic numerical service
  ↓
Verified result
  ↓
Local LFM explanation

```

Appropriate responsibilities include:

- vendor-field mapping proposals;
- natural-language → Position DSL;
- natural-language → Strategy DSL;
- experiment configuration;
- claim/evidence search;
- numerical-output explanation;

- report generation.

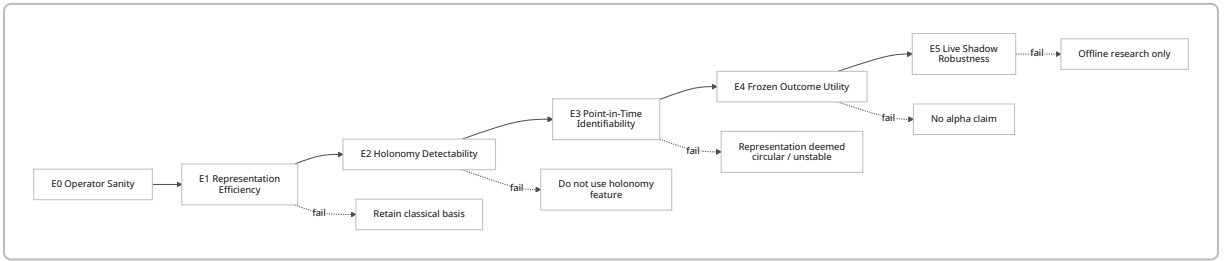
Prohibited responsibilities include:

- inventing option prices or Greeks;
- deciding whether data leakage is acceptable;
- silently changing a frozen split;
- fabricating unavailable historical contracts;
- interpreting missing values as observations;
- placing live orders;
- assigning an evidence status without a corresponding numerical gate.

Official LFM2.5 GGUF repositories make local deployment technically plausible on modest hardware; exact project rights and commercial-license compatibility should remain explicitly unresolved until reviewed. 5

Validation and verification

E0 should remain the numerical prerequisite, but E1–E5 form the scientific commercialization ladder.



Gate E1 — representation efficiency

The prior synthetic result should be reclassified as **E1-A: oracle calibration** because the clean sections were generated using the same low connection-Laplacian eigenspace that was subsequently used for reconstruction.

At the reported $k = 6$,

$$D_{\text{conn}} = 0.020817, \quad D_{\text{PCA}} = 0.022738, \quad D_{\text{DCT}} = 0.025206.$$

The conventional MSE reduction relative to DCT is

$$\frac{0.025206 - 0.020817}{0.025206} \approx 17.4\%.$$

The reverse denominator gives the separate statement that DCT incurred about 21.1% more error than the connection basis.

A production E1 should contain the following comparisons:

Basis	Learns from signals?	Uses graph?	Uses transport?	Gauge-fair role
Raw DCT	No	No	No	Naïve temporal baseline
Gauge-DCT	No	Path/tree	Alignment only	Essential fair baseline
Scalar graph Fourier	No	Yes	Identity	Tests topology without connection
Finite-sample PCA	Yes	No	No	Classical learned benchmark
Population/oracle PCA	Oracle	No	No	Synthetic upper reference
Connection SFT	No signal training if G, U given	Yes	Yes	Candidate structural basis
Wrong/shuffled connection	No	Yes	Wrong	Negative control
Random orthogonal	No	No	No	Sanity floor

For each independently generated synthetic world g , define a preregistered difference

$$\Delta_g(R_*) = D_{\text{baseline},g}(R_*) - D_{\text{conn},g}(R_*).$$

The principal inferential unit should be the world—not each noisy section generated inside it.

A robust confirmatory design is:

- at least a pilot-sized set of independent worlds sufficient to estimate the variance of Δ_g ;
- choose final G by prospective power analysis;
- freeze one target rate R_* before evaluating final worlds;
- report paired bootstrap confidence intervals;
- use a paired permutation test or preregistered paired test across worlds;
- correct the family if several primary comparators are tested.

Power should be simulated under controlled advantages

$$\delta = \mathbb{E}[\Delta_g]$$

and final G chosen to achieve a declared target such as 80% or 90% power at the selected significance level rather than choosing an arbitrary number of worlds.

E1 also requires parameter sweeps over:

σ_{noise} , $\sigma_{\text{transport}}$, false-edge rate, missing-edge rate, ϕ_{holonomy} , n_{PCA} , R .

A legitimate prior should lose its advantage as the supplied graph/connection becomes wrong.

Gate E2 — holonomy null gate

E2 tests whether the implementation can distinguish true planted cycle inconsistency from transport-estimation noise.

The primary null should be a noisy but globally integrable connection:

$$H_0 : U_{ij} = G_j^\top G_i \exp(\eta_{ij}),$$

where η_{ij} follows the calibrated estimation-noise law.

The alternative plants nontrivial cycle holonomy:

$$H_1 : H_c = R(\phi_c), \quad \phi_c \neq 0. \quad =$$

The null must preserve:

- graph topology;
- cycle lengths;
- degree distribution;
- edge weights;
- transport-noise scale;
- frame-estimator uncertainty.

A generic GUE null has no special justification here; the uploaded studio blueprint similarly argues that null models must preserve the structure relevant to the domain. [filecite turn0file1](#)

Possible E2 statistics are:

$$T_{\text{angle}} = |\phi_c|,$$

$$T_{\text{trace}} = 2 - \text{tr}(H_c),$$

$$T_F = \|I - H_c\|_F,$$

and global spectral frustration such as

$$T_\lambda = \lambda_{\min}(L_U).$$

For many cycles, use a maximum statistic

$$T_{\max} = \max_{c \in \mathcal{C}} Z_c$$

and calibrate the **maximum** under each null replicate.

With B null replicates,

$$p_{\text{emp}} = \frac{1 + \#\{T_b \geq T_{\text{obs}}\}}{B + 1}.$$

This exact plus-one Monte Carlo convention avoids a reported zero probability.

The power experiment should sweep

$$|\phi| \times \sigma_{\text{transport}} \times L_{\text{cycle}}$$

and produce

$$\Pr(\text{detect} \mid \phi, \sigma, L).$$

A valid E2 pass means:

- type-I error is calibrated near the declared level;
- power increases with planted holonomy magnitude;
- power falls gracefully with transport noise;
- statistics remain invariant under random vertex gauges;
- the result replicates across independently generated graphs.

It does **not** establish profitable cyclic trading.

Gate E3 — point-in-time identifiability

E3 is arguably the most important gate for applying the geometry to markets.

Its first component is an exact causal-software certificate:

Editing, deleting, or randomizing every event with availability time after cutoff T must leave every feature, graph edge, transport, strategy state, and model input dated at or before T unchanged.

This can be tested by future-data poisoning followed by artifact-hash comparison.

The second component asks whether the learned graph and transports encode anything beyond the variables used to construct them.

A powerful design is **cross-family holdout**:

construct graph/connection from feature family A ,

then test

alignment or prediction of completely held-out family B .

Examples:

Construction family	Held-out audit family
Returns + technicals	Option-surface movement
Price + volume	Order-flow imbalance
Vol surface	Future realized-volatility state
Slow condition state	Fast activity state

The graph is useful as a representation only if its structure generalizes beyond the features that directly defined its distances.

The comparable nulls should preserve marginal geometry while breaking specific structure:

- similarity-edge endpoint shuffle preserving degree;
- transport permutation across matched edges;
- random orthogonal transport with the same noise magnitude;
- chronological-only graph;
- identity connection;
- feature-family permutation within matched regime blocks.

The uploaded zeta/studio work makes this “inserted versus independently recovered” distinction central to the product thesis. [filecite turn0file1](#)

E3 should additionally measure stability across neighboring training periods:

$$\|P_A - P_B\|_F$$

for low spectral subspace projectors,

$$J(E_A, E_B) = \frac{|E_A \cap E_B|}{|E_A \cup E_B|}$$

for graph-edge survival, and distributional stability for holonomy and TDA summaries.

Failure of E3 means the geometry remains a descriptive visualization rather than a validated feature representation.

Gate E4 — frozen predictive utility

E4 finally asks whether advanced representation features improve a prespecified **future position-outcome task**.

The task should be distributional. For example:

$$Y_{t,h}(P) = [\text{net PnL}, \text{MFE}, \text{MAE}, \text{drawdown}, \mathbf{1}_{\text{stop}}, \mathbf{1}_{\text{target}}].$$

The model ladder should be frozen in advance:

Model	Input
Unconditional	Position family + horizon only
Raw analog	Canonical raw state
Linear/regularized	Raw state + position
Tree/boosting	Raw state + position
PCA analog/model	PCA state coordinates
Gauge-DCT/GFT	Gauge-fair spectral coordinates
Connection SFT	Connection spectral features
Raw + TDA	Adds persistence
Raw + SFT + TDA	Combined geometric candidate
MoE	Only after prior levels justify complexity

Primary distributional metrics can include weighted quantile/pinball loss and CRPS-like distributional loss; binary event heads can use Brier/log loss. Calibration must be shown separately from discrimination.

For each test block b ,

$$\Delta_b = L_{\text{baseline},b} - L_{\text{candidate},b}.$$

Use a block or hierarchical bootstrap at the level of economically meaningful time/instrument blocks rather than treating every overlapping bar as independent.

The data should be divided into:

development → quarantine → frozen confirmation.

Overlapping labels should be purged from adjacent train/test boundaries. The uploaded options synthesis specifically recommends frozen protocol, purging/embargo, honest trial accounting, and disjoint-window replication for this reason. [filecite turn0file2](#)

If the workstation searches many strategies or model variants, every attempted configuration enters the trials ledger. Selection-adjusted performance can then be cross-checked with the Deflated Sharpe Ratio and Probability of Backtest Overfitting framework. DSR explicitly addresses selection bias, backtest overfitting, and non-normality; PBO/CSCV estimates how often an in-sample-optimal configuration performs poorly out of sample. ⁴

A strategy-level “alpha” claim should require all of the following:

incremental held-out loss improvement
 +economic improvement after declared costs
 +acceptable calibration
 +multiple-trial correction
 +replication across independent regimes.

A visually separated terrain is not enough.

Gate E5 — live shadow robustness

E5 runs the exact frozen model and feature pipeline against live data **without order authority**.

Its primary concern is research-to-production parity.

The workstation should record:

historical replay output $\stackrel{?}{=}$ live-equivalent replay output

when supplied the same canonical event stream.

Metrics include:

- deterministic feature parity;
- event loss and duplication;
- late-event rate;
- p50/p95/p99 processing latency;
- missing-chain frequency;
- model abstention frequency;
- terrain effective sample size;
- position-transport failure rate;
- calibration by week/month;
- representation drift;
- distributional-loss drift;
- strategy-state parity;
- model version consistency.

Thresholds should be deployment-specific and frozen before E5 rather than invented after observing performance.

A sequential failure detector may monitor whether the distribution of

$$\Delta L_t = L_{\text{candidate},t} - L_{\text{baseline},t}$$

moves outside the expected E4 range.

E5 failure means the model returns to offline or shadow research even if E4 passed historically.

Required V&V artifact set

Artifact	Required contents
Dataset manifest	Source, contract, timestamps, revisions, checksum
Coordinate manifest	Semantics and units of every coordinate
Feature DAG	Formula/version/availability rule for each feature
Graph bundle	Vertices, edges, reasons, weights, cutoff
Connection bundle	Frames, transports, uncertainties, group
Basis bundle	Eigenvalues, projectors, solver/version
TDA bundle	Metric, window, filtration, PH backend
Position DSL	Legs/template/valuation convention
Position-transport manifest	Match objective, residuals, failures
Strategy DSL	All automaton predicates
Split manifest	Development/quarantine/frozen blocks
Trial ledger	Every configuration evaluated
Null manifest	Generator, justification, seeds
Run environment	Code commit, packages, CPU/GPU, model revision
Residual table	Per-world/per-block primary statistics
Evidence ledger	Claim → gate → result → limitations

Implementation roadmap, compute, datasets, and reproducible demos

The implementation should deliberately begin with a product that remains useful even if topology and sheaf features fail E4.

Prioritized build sequence

Priority	Deliverable	Why first
P0	Event store + coordinate registry	Everything depends on point-in-time correctness
P0	Position DSL + deterministic valuation	Defines the query object
P0	Historical position transport	Required for arbitrary-position terrain

Priority	Deliverable	Why first
P0	Replay clock + strategy automaton	Makes every rule reproducible
P0	Raw analog terrain	Establishes interpretable baseline
P1	Evidence ledger + trial registry	Prevents research debt
P1	Graph/frame/connection kernel	Enables E0/E1/E2
P1	Gauge-fair comparator suite	Prevents false geometric victories
P1	Connection/sheaf spectral basis	Candidate representation
P1	TDA pipeline	Candidate structural features
P2	Distributional baseline models	E4 evaluation
P2	Microstructure encoder	Adds high-frequency activity state
P2	Local LFM semantic sidecar	Improves workflow without touching numerical truth
P3	Mixture of experts	Only after simpler models pass
P3	Live shadow mode	Requires frozen offline pipeline
P4	Execution integration	Separate later product decision

The correct product sequence is therefore not “build the neural network first.” It is:

data truth → position truth → replay truth → baseline terrain → advanced geometry → ML.

Software architecture

The uploaded commercialization blueprint already proposes a sensible separation between a TypeScript browser client, Python numerical service, asynchronous computation, object storage, and run metadata.   The market version can extend that concept:

```
apps/
  workstation-web/

services/
  api/
  replay-engine/
  numerical-worker/
  model-worker/
  lfm-local/

packages/
  schemas/
```

```
position-dsl/  
strategy-dsl/  
evidence/  
market-data/  
geometry/  
topology/  
terrain/  
validation/  
  
plugins/  
  indicators/  
  option-transport/  
  microstructure/  
  null-models/  
  models/  
  data-adapters/
```

A practical numerical core is sparse linear algebra plus tensor operations; the low connection eigenproblem scales with the block Laplacian dimension nr .

For graph degree $\bar{d}$,

$$|E| \approx \frac{n\bar{d}}{2},$$

and block-sparse storage scales approximately as

$$O(|E|r^2).$$

This is why stalk rank should remain modest in the first version.

For example, $r = 4\text{--}16$ is a more credible initial range than attempting hundreds of coordinates inside each stalk. Hundreds of raw indicators can first be compressed within their feature families before connection construction.

Persistent homology should similarly operate on bounded sliding-window clouds or landmarks rather than unrestricted entire market histories. The mathematical value is in time-local structural descriptors, not in constructing the largest possible Rips complex.

Local LFM compute

A local semantic agent need not consume the same GPU as the numerical training stack. Official LFM2.5 GGUF listings make a small separate process feasible: the 1.2B Instruct Q4_K_M artifact is listed around 731 MB, whereas the larger 8B-A1B Q4_K_M artifact is about 5.16 GB. ⁵

A sensible configuration is:

```

semantic_model:
  role: schema_mapper_and_analyst
  model_repo: pinned
  revision: pinned_commit
  tokenizer_hash: sha256
  quantization: pinned
  context_policy: local
  numerical_authority: false

```

“Latest” may be used only as a developer discovery channel. Published experiments must pin exact revisions.

Model and tokenizer licenses, derivative-work terms, exchange-data redistribution rights, and option-market-data licenses remain unresolved assumptions until contract/legal review.

Market-data scale

For a market-facing options workstation, data cost and storage can dominate the initial engineering more than the connection spectral calculation.

Cboe's current official Option Quote Intervals product offers listed U.S. option quotes from 1-minute or custom intervals with NBBO fields and optional implied-volatility/Greek calculations. Cboe reports historical availability from 2012 on the product page and estimates roughly **20 GB compressed per year for SPY at one-minute intervals with calculations, 45 GB for SPX**, and approximately **2.3 TB compressed per year for the full U.S. listed-option universe** at the same interval/calculation configuration. ¹⁸

That strongly argues for a narrow first dataset.

Dataset	First use	Scope	Access assumption
Synthetic bundle worlds	E1/E2	Unlimited generated	Available
Synthetic quasiperiodic phase data	Mathematical reference	Small	Available
Cboe option intervals	Position terrain	One liquid ETF first	License/access unknown
LOBSTER reconstructed book	Microstructure research	Selected NASDAQ equities	License/access unknown
Live option feed	E5 shadow	Same contract schema	Not assumed
Semantic event feed	Catalyst layer	Optional	Not assumed

LOBSTER's official site describes reconstructed limit-order-book datasets for research/professional analysis and a current platform that supports high-frequency order-book reconstruction. Access terms for this project should nevertheless be treated as unknown until actually obtained. ¹⁹

A realistic local research target is therefore **one underlying or a very small liquid universe**, not the full listed options market.

For SPY one-minute option data, a planning allocation of approximately 2–3× the raw compressed dataset size per year is reasonable for derived Parquet partitions, indices, manifests, position-replay results, and temporary artifacts; that multiplier is an engineering estimate, not a vendor specification.

Minimal compute envelope

A credible local development machine can begin with:

Resource	MVP planning target
CPU	16–32 modern cores
RAM	64 GB minimum; 128 GB preferable for larger state graphs
GPU	Optional 12–24 GB VRAM for learned models
Fast local storage	1–2 TB NVMe
Archival storage	Expand with dataset scope
Local semantic model	1.2B quantized class initially
Sparse spectral backend	CPU correctness reference first
Null simulations	Parallel CPU workers

These are engineering recommendations rather than measured requirements. Actual capacity should be derived from E1/E2 benchmark manifests.

The full-market Cboe storage figures make clear that production-scale multi-underlying intraday option research will eventually require substantially more storage than the minimal local system. ¹⁸

Minimal reproducible mathematics demo

The first reference demo should be deliberately outside finance.

Define

$$\Phi(t)_q = a_q e^{-it\omega_q},$$

with declared frequencies such as

$$\omega_q = \log q.$$

Then

$$\|\Phi(t) - \Phi(u)\|^2 = 2 \sum_q a_q^2 [1 - \cos((t - u)\omega_q)].$$

This gives an exact stationary distance kernel.

The demo should show:

- one-dimensional parameterized trajectory;
- higher-dimensional closure for independent frequencies;
- exact distance-vs-gap identity;
- local tangent rank;
- recurrence neighbors versus parameter-local neighbors;
- graph connection;
- random-gauge invariance;
- E1-A oracle compression;
- E1-B gauge-DCT;
- E2 planted holonomy.

The purpose is not to claim anything new about zeta zeros. It is a CI-visible mathematical certificate demonstrating the same representation-versus-discovery discipline emphasized by the uploaded scientific-studio blueprint. [filecite turn0file1](#) Quasiperiodic dense-torus behavior also has an independent theoretical foundation in the sliding-window literature. [17](#)

Minimal reproducible market demo

The first market demo should use one liquid underlying, preferably an ETF for which synchronized underlying data is straightforward.

A concrete demo specification is:

```
Underlying: one liquid ETF
Options: 1-minute historical chain snapshots
History: enough for separate development / quarantine / confirmation periods
Position types:
    long call
    vertical spread
    straddle
    iron condor
    user-defined arbitrary multi-leg

Horizons:
    30 minutes
    session close
    5 sessions
    expiration where applicable
```

State families:
returns
realized volatility
volume
selected technical indicators
option surface
liquidity
calendar/session

Advanced families:
connection spectrum
persistent topology
optional microstructure

Terrain:
weighted analog distribution
PnL quantiles
MFE / MAE
transport quality
analog count
calibration status

The first market demonstration should **not** require a neural network.

The success criterion for the platform MVP is that a user can build any supported position, move the replay clock to a historical instant, transport the position across comparable historical chain fibers, retrieve actual analog outcomes, and see exactly why those analogs were selected.

Only then should connection/TDA features be added and evaluated to determine whether the analog set or prediction loss improves.

Roadmap

Window	Main objective	Gate target	Concrete deliverable
Months 0–2	Point-in-time core	E0	Event store, coordinate registry, DSLs
Months 2–4	Position Terrain baseline	Pre-E4 baseline	Replay + position transport + analogs
Months 4–6	Geometry kernel	E1	Connection/SFT + gauge comparator
Months 6–7	Holonomy research	E2	Null/power dashboard
Months 7–9	Causal geometry	E3	Train-only graph, cross-family holdout
Months 9–12	Outcome modeling	E4	Distributional models + frozen confirmation

Window	Main objective	Gate target	Concrete deliverable
Months 12–14	Local semantic UX	Supporting	LFM schema/compiler integration
Months 14–16	Streaming shadow	E5	Live/replay parity
Later	Execution/advisory	Separate program	Not part of core validation

Workstation visualization, provenance, and final recommendation

The UI should feel like a professional market workstation, but its differentiator should be **scientific inspectability** rather than the number of charts visible at once.

The uploaded GT-MoE visualization concept combines live signals, a geometric trajectory, persistence information, regime state, and expert activity on one operational screen. That layout principle transfers naturally to markets. [filecite turn0file3](#)

Position and chain panel

The left-side position workspace should show:

- complete leg table;
- live/replay bid/ask;
- mark convention;
- position Greeks;
- scenario PnL;
- historical transport mode;
- transport residual;
- liquidity quality;
- position hash.

The payoff diagram should be only one view. The more important chart is how the position's exposure state evolves with spot, volatility, and time.

Market timeline

A synchronized chart should display:

```
price
volume
volatility
declared indicators
surface events
semantic events
```

strategy transitions
position actions

During replay, future information must be physically inaccessible to the client state rather than merely hidden graphically.

Position Terrain panel

The central terrain should be selectable by output:

median PnL, $q_{0.05}$, $q_{0.95}$, MAE, MFE, $\Pr(\text{stop})$, N_{eff} .

Every displayed region should expose both outcome and support.

A useful visual convention is:

high confidence / dense history	→ fully rendered terrain
low effective sample size	→ faded terrain
out-of-distribution	→ crosshatched / UNKNOWN
failed position transport	→ absent
failed evidence gate	→ disabled analytical layer

The workstation should never fill sparse regions with visually smooth certainty without visibly labeling the interpolation.

Analog explorer

Clicking any terrain point should reveal the actual historical analogs:

Analog field	Why it matters
Date/time	Detect clustering in one regime
State distance	Understand similarity
Position transport	Verify economic comparability
Transport residual	Detect poor match
Subsequent path	See distribution rather than median
Costs	Prevent theoretical-PnL bias
Regime label	Show concentration
Feature attribution	Explain distance
Evidence scope	Distinguish descriptive/validated

This may ultimately become more valuable to expert users than a neural-model explanation.

Geometry and topology inspector

A dedicated scientific panel should show:

- graph edges by type;
- chronological versus recurrence edges;
- local frames;
- transport matrices;
- connection energy;
- low spectrum;
- gauge-DCT comparison;
- cycle holonomy;
- persistence diagram;
- Betti curves;
- representation ablations.

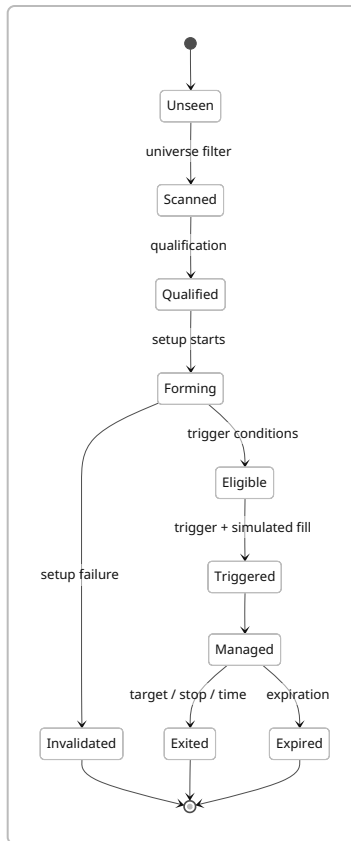
A single toggle should perform a **random gauge transformation** and demonstrate that invariant quantities remain unchanged.

Another should disable similarity loops and show whether the claimed spectral benefit survives.

These are far more informative than a static “3D manifold” animation.

Strategy grammar panel

The policy should be visible as an automaton:



Clicking a transition should expose the exact predicate, required data fields, and value at that timestamp.

V&V dashboard

The most important top-level status should be a matrix rather than a single “validated” badge.

Gate	Question	Example status
E0	Is the numerical operator correct?	Certified
E1	Does the basis earn compression complexity?	Pending
E2	Is holonomy detectable above noise?	Pending
E3	Is the representation noncircular and causal?	Pending
E4	Does it improve frozen position outcomes?	Not tested
E5	Does it survive live shadow conditions?	Not tested

A candidate method can be useful even if only E0–E3 pass; it simply remains a descriptive/research representation.

Trial and selection dashboard

Because strategy/model search is itself a multiple-testing process, every run must remain visible.

The dashboard should show:

$N_{\text{attempted}}$, $N_{\text{effective}}$, best IS, best OOS, DSR, PBO, frozen replication.

The prior neuroevolution/options analysis is particularly relevant here: the optimizer's ability to search a large strategy space is not itself an advantage if the finite historical path cannot support that number of independent decisions. [filecite5turn0file25](#) DSR and PBO were developed precisely to address selection and backtest-overfitting concerns. [4](#)

Local semantic command center

The local LFM can provide a command interface such as:

"Build a 30-DTE delta-neutral straddle template and compare its five-session terrain when implied-versus-realized volatility is in the top quartile, but do not use future earnings information."

The LFM output should be rendered first as proposed typed configuration:

```
position:
  template: straddle
  tenor_target_days: 30
  delta_target: neutral

condition:
  iv_minus_rv_quantile:
    min: 0.75

horizon:
  sessions: 5

information_policy:
  point_in_time: true
  future_events: forbidden
```

The user approves or edits the specification; deterministic software executes it.

That architecture preserves the convenience of a local semantic model without allowing language-model hallucination to become market data.

Provenance panel

Every plot should have a “Why does this exist?” action that yields a manifest similar to:

```
artifact:
  id: terrain_9b123
  type: position_outcome_terrain

dataset:
  hash: sha256:...
  vendor_contract: internal_reference
  point_in_time_policy: availability_time

position:
  dsl_hash: sha256:...
  transport_version: exposure_transport_v1

representation:
  feature_dag_hash: sha256:...
  graph_hash: sha256:...
  connection_hash: sha256:...
  spectral_basis_hash: sha256:...
  tda_config_hash: sha256:...

experiment:
  split_hash: sha256:...
  trial_registry_hash: sha256:...
  null_manifest_hash: sha256:...

software:
  git_commit: ...
  environment_hash: ...
  numerical_backend: ...
  random_seed_tree: ...

semantic_layer:
  model_repo: ...
  revision: ...
  quantization: ...
  tokenizer_hash: ...
  numerical_authority: false

evidence:
  status: DESCRIPTIVE
  gate_scope:
    E0: PASS
    E1: PASS
```

E2: PASS
E3: PASS
E4: NOT_RUN
E5: NOT_RUN

This is where the workstation becomes something substantially different from a conventional trading dashboard.

Recommended research priority

The source hierarchy for implementation should be:

Priority	Research basis	Role
Highest	Singer & Wu connection Laplacian ²⁰	Frame transport and vector spectral geometry
Highest	Hansen & Ghrist cellular sheaves ¹¹	Sheaf/coboundary/Laplacian semantics
Highest	D'Acunto et al. 2026 SFT preprint ¹²	Modern sheaf-Fourier/filtering formulation
Highest	Shuman et al. graph signal processing ²¹	Classical GFT comparator
Highest	Bandeira–Singer–Spielman synchronization ²²	Connection/frustration validation
Highest	Gao–Brodzki–Mukherjee synchronization geometry ¹³	Gauge/holonomy interpretation
High	Perea–Harer sliding-window persistence ¹⁶	TDA of temporal signals
High	Gakhar–Perea quasiperiodic persistence ¹⁷	Orbit/closure discipline
High	Cont–Kukanov–Stoikov OFI ²³	Microstructure baseline
High	Cont–Stoikov–Talreja LOB model ²⁴	State-conditioned execution/activity model
Comparator	DeepLOB ⁷	Learned microstructure benchmark
Validation	Bailey–López de Prado DSR ²⁵	Trial-selection correction
Validation	Bailey et al. PBO/CSCV ²⁶	Backtest-overfitting audit
Hedging benchmark	Bühler et al. Deep Hedging ²⁷	Well-posed derivative-policy comparison

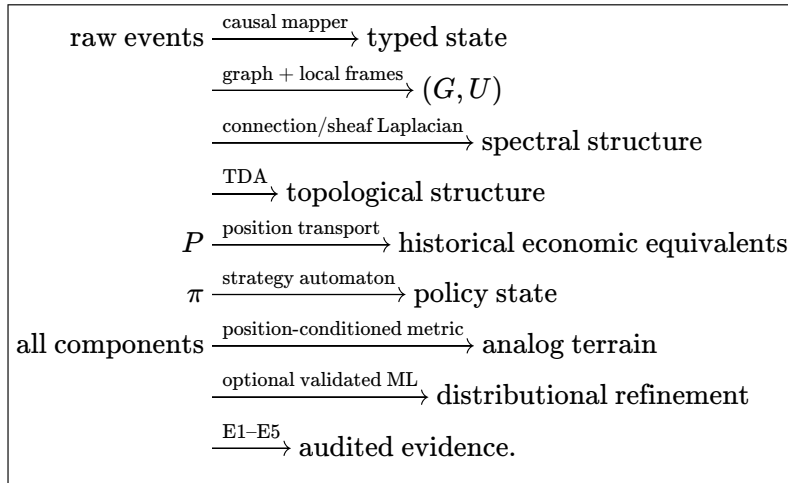
Priority	Research basis	Role
Data	Cboe DataShop official options data ¹⁸	Historical option chains
Microstructure data	LOBSTER official platform ¹⁹	Historical order books
Semantic layer	LiquidAI LFM2.5 model repositories ⁵	Local semantic compiler

Final algorithm of merit

The resulting algorithm should be defined operationally as:

$$\text{PCT-SFT}(P_t, \pi, h) = \left[\widehat{\mathcal{L}}(Y_{t,h} \mid Z_t(P_t, \pi)), \right. \\ \mathcal{N}_t(P_t, \pi), \\ N_{\text{eff}}, \\ \text{transport uncertainty}, \\ \text{OOD status}, \\ \left. \text{evidence status} \right].$$

Its construction is:



The key innovation is therefore **not any one Laplacian, TDA algorithm, neural network, or technical indicator**. All of those have substantial prior literature.

The potentially distinctive contribution is the integration of:

economic position transport
 +**gauge-audited connection geometry**
 +**sheaf/graph spectral representations**
 +**persistent topology**
 +**position-conditioned historical similarity**
 +**formal strategy grammar**
 +**microstructure-aware activity state**
 +**local semantic compilation**
 +**immutable statistical evidence governance.**

The uploaded scientific-studio blueprint already points toward “geometry with epistemic provenance”; the market implementation makes the **position itself** part of that geometry. [filecite turn0file1](#) The Mario project contributes the useful principle that a controller should react to a structured world-state rather than memorize an input sequence, while the options synthesis supplies the necessary correction that markets do not offer unlimited independent worlds for optimization. [filecite turn0file0](#) [filecite turn0file2](#)

That yields the most defensible market-facing product thesis:

The Position Terrain Workstation does not ask whether a chart pattern “works.” It asks: for this exact economic exposure, under this point-in-time market condition, activity state, strategy state, transport rule, and horizon, what happened in genuinely comparable historical states; what geometry was imposed to find those states; what independent evidence supports that geometry; how uncertain is the resulting outcome distribution; and did the advanced representation outperform the simpler one on data that was never used to choose it?

That is a viable algorithm of merit—and, crucially, one that can still produce a scientifically useful negative answer.

1 21 The Emerging Field of Signal Processing on Graphs: Extending High-Dimensional Data Analysis to Networks and Other Irregular Domains

https://arxiv.org/abs/1211.0053?utm_source=chatgpt.com

2 16 Sliding Windows and Persistence: An Application of Topological Methods to Signal Analysis

https://arxiv.org/abs/1307.6188?utm_source=chatgpt.com

3 22 A Cheeger Inequality for the Graph Connection Laplacian

https://arxiv.org/abs/1204.3873?utm_source=chatgpt.com

4 25 The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality by David H. Bailey, Marcos Lopez de Prado :: SSRN

https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID2460551_code87814.pdf?abstractid=2460551&utm_source=chatgpt.com

5 LiquidAI/LFM2.5-1.2B-Instruct-GGUF · Hugging Face

https://huggingface.co/LiquidAI/LFM2.5-1.2B-Instruct-GGUF?utm_source=chatgpt.com

6 23 The Price Impact of Order Book Events

https://arxiv.org/abs/1011.6402?utm_source=chatgpt.com

7 DeepLOB: Deep Convolutional Neural Networks for Limit Order Books

https://arxiv.org/abs/1808.03668?utm_source=chatgpt.com

8 Deep Sets

https://arxiv.org/abs/1703.06114?utm_source=chatgpt.com

9 Conformalized Quantile Regression

https://arxiv.org/abs/1905.03222?utm_source=chatgpt.com

10 20 Vector Diffusion Maps and the Connection Laplacian

https://arxiv.org/abs/1102.0075?utm_source=chatgpt.com

11 Toward a Spectral Theory of Cellular Sheaves

https://arxiv.org/abs/1808.01513?utm_source=chatgpt.com

12 Sheaf-theoretic Signal Processing on Graphs: Spectral Theory, Filtering, and Sampling

https://arxiv.org/abs/2608.01318?utm_source=chatgpt.com

13 The Geometry of Synchronization Problems and Learning Group Actions

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14 Cycle holonomy induces higher-order constraints and controls remote synchronization transitions via twisted Laplacian spectra

https://arxiv.org/abs/2604.19682?utm_source=chatgpt.com

15 Attribute Compression of 3D Point Clouds Using Laplacian Sparsity Optimized Graph Transform

https://arxiv.org/abs/1710.03532?utm_source=chatgpt.com

17 Sliding Window Persistence of Quasiperiodic Functions

https://arxiv.org/abs/2103.04540?utm_source=chatgpt.com

18 Option Quotes

https://datashop.cboe.com/option-quote-intervals?utm_source=chatgpt.com

19 LOBSTER | high quality limit order book data.

https://data.lobsterdata.com/?utm_source=chatgpt.com

24 A Stochastic Model for Order Book Dynamics | Operations Research

https://pubsonline.informs.org/doi/10.1287/opre.1090.0780?utm_source=chatgpt.com

26 The probability of backtest overfitting - Journal of Computational Finance

https://www.risk.net/ja/node/2471206?utm_source=chatgpt.com

27 Deep Hedging

https://arxiv.org/abs/1802.03042?utm_source=chatgpt.com